# Install the Cloudflare Enforcer

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If your organization uses **Cloudflare,** you can use HUMAN's Cloudflare **Enforcer** to protect against malicious behavior. The Cloudflare Enforcer is installed using a Cloudflare **Worker,** or a snippet of code, and is deployed to your content delivery network (CDN). The Enforcer dictates how traffic should be handled per your organization's standards.

You can learn how to install the Cloudflare Enforcer with this article.

## Prerequisites

- A **Cloudflare account.** You can sign up for an account from **Cloudflare's website.**
- A **command-line interface (CLI)**.
- A text editor.
- **Node Version Manager (nvm)** installed on your device. See nvm's **GitHub repository** to learn how to install it.
- The latest version of **Node.js.** After installing nvm, enter `nvm install stable` in your CLI to install it.
- **Wrangler** installed on your device. You can enter `npm install -g wrangler` in your CLI to install it.
- Your unique HUMAN information:
  - Your **Application ID.** You can find this under **Platform Settings > Applications > Overview** in the HUMAN console. If you have multiple environments, you will also have multiple Application IDs, so be sure to choose the correct ID for the environment you want to install on.
  - Your **Server Token.** You can find this under **Platform Settings > Applications > Status & Settings > Server Token.**
  - Your **Risk Cookie Key.** You can find this under **Bot Defender > Policies > Policy Settings > Policy Information.**

# Installation

There are two parts to the Cloudflare Enforcer installation:

1. **Create the Cloudflare worker**
2. **Install the HUMAN Enforcer**

The majority of the installation process is done through a CLI.

## Create the Cloudflare Worker

1. In your CLI, enter `npm create cloudflare@latest`. If you're prompted to install additional packages to complete the installation, hit `y` to do so.

Shell Copy

```
Need to install the following packages:
create-cloudflare@2.XX.Y
Ok to proceed? y/n
```

2. When prompted, enter a folder name to create your Worker in. This will also be the Worker's name in your Cloudflare dashboard. In the example below, we've named our folder `HUMAN-cloudflare-enforcer`.

Shell Copy

```
npm create cloudflare@latest

using create-cloudflare version 2.XX.Y
┌ Create an application with Cloudflare Step 1 of 3
│
└ In which directory do you want to create your application? also use
  ./HUMAN-cloudflare-enforcer
```

3. When prompted, select the `"Hello World" Worker` option and hit `return` to proceed.

Shell Copy

```
│ dir ./HUMAN-cloudflare-enforcer
│
└ What type of application do you want to create?
  ● "Hello World" Worker
  ○ "Hello World" Worker (Python)
  ○ "Hello World" Durable Object
  ○ Website or web app
  ○ Example router & proxy Worker
  ○ Scheduled Worker (Cron Trigger)
  ○ Queue consumer & producer Worker
  ○ API starter (OpenAPI compliant)
  ○ Worker built from a template hosted in a git repository
```

4. Select whether to install the Worker using TypeScript. By default, the Worker will install in JavaScript.

Note

Whether you use TypeScript or JavaScript is up to you, but be sure to **remember what you choose.** Later in the installation, you will need to choose the code snippet that corresponds to the language you installed in.

Shell Copy

```
├ What type of application do you want to create?
│ type "Hello World" Worker
│
└ Do you want to use TypeScript?
  Yes / No
```

5. When prompted, select `No` to deploying your application.

Shell Copy

```
┌ Deploy with Cloudflare Step 3 of 3
│
└ Do you want to deploy your application?
  Yes / No
```

Afterwards, you should receive a success message that reads `APPLICATION CREATED`. This means you created a basic Cloudflare Worker, and you're ready to move on to installing the HUMAN Enforcer.

## Install the HUMAN Enforcer

1. In your CLI, enter `cd folder_name` to open the folder you created in **Create the Cloudflare Worker, Step 2.** Based on our earlier example, we would enter `cd HUMAN-cloudflare-enforcer`.
2. Enter `npm i --save @humansecurity/cloudflare-enforcer`. This will install the latest Cloudflare Enforcer from HUMAN.
3. Enter `cd src`.
4. In your preferred text editor, open either `index.ts` or `index.js` depending on your choice in **Create the Cloudflare Worker, Step 4**. The `.ts` file is for TypeScript installations, while `.js` is for JavaScript.

> Note
>
> You can check the file type you have by entering `ls` while inside the `src` folder. This will show you your `index` file's type.

5. Delete all the code currently in the `index` file. The following snippet shows all the default code you should delete.

Shell Copy

```
/**
 * Welcome to Cloudflare Workers! This is your first worker.
 *
 * - Run `npm run dev` in your terminal to start a development server
 * - Open a browser tab at http://localhost:8787/ to see your worker
 * - Run `npm run deploy` to publish your worker
 *
 * Learn more at https://developers.cloudflare.com/workers/
 */
export default {
        async fetch(request, env, ctx) {
                return new Response('Hello World!');
        },
};
```

6. Copy the appropriate code snippet, either TypeScript or JavaScript, from below and paste it into your `index` file.

Note

In addition to choosing between TypeScript and JavaScript, you must also make sure to pick the right type of Worker. This can either be an ES Module or Service Module, and it depends on your Cloudflare configuration. You can see some examples from both Workers in **Cloudflare's documentation.**

ES Modules Syntax: JavaScript

ES Modules Syntax: TypeScript

Service Worker Syntax: JavaScript

Service Worker Syntax: TypeScript

7. Update the `px_app_id`, `px_auth_token`, and `px_cookie_secret` fields with your **Application ID, Server Token,** and **Risk Cookie Key** respectively.

> **Note**
>
> These are the **minimum** required fields that **must be completed** in order for the Enforcer to work. You can always return to this file to customize your Enforcer later with our **optional configurations.**

Shell | Copy

```shell
// define an enforcer configuration however you see fit
const config = {
    px_app_id: '<APP_ID>',
    px_auth_token: '<AUTH_TOKEN>',
    px_cookie_secret: '<COOKIE_SECRET>',
    // ...
};
```

8. Save and close the file.
9. Enter `npx wrangler deploy` to deploy your Worker. You may be prompted to log in to your Wrangler or Cloudflare account.
10. Navigate to your Cloudflare dashboard and open **Workers & Pages.** Your new Worker with the HUMAN Enforcer should appear with the same name you gave it in **Create the Cloudflare Worker, Step 2.** In our case, the Worker is named **HUMAN-cloudflare-enforcer.**
11. Select the **Worker,** then select **Settings > Triggers > Add route** under **Routes**. This is where you can you add URL routes and zones to monitor with the Cloudflare Enforcer. We recommend protecting your full domain, including all pages on your domain, with the pattern `subdomain.apex.com/*` . For example, to protect the full domain of a site with the url `www.example.com` , you would provide the following fields:
    - **Route:** `www.example.com/*`
    - **Zone:** `example.com`

Your Cloudflare Enforcer has been successfully installed with the minimum requirements to monitor activity on your Cloudflare CDN. You can further customize the Enforcer's behavior by referencing our **configuration options.**

Once you finish installing, be sure to contact HUMAN to complete your tuning process.

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